UNIVERSITY OF GHANA

DEPARTMENT OF STATISTICS & ACTUARIAL SCIENCE

B.SC./B.A. SECOND SEMESTER EXAMINATIONS: 2018/2019

STAT 112: ELEMENTARY PROBABILITY (3 Credits)

INSTRUCTIONS: ANSWER ALL QUESTIONS IN SECTION A

AND ANY TWO FROM SECTION B

(Statistical tables are to be provided)

TIME: TWO AND A HALF HOURS

SECTION A (50 MARKS)

Answer ALL Questions

A1. (a) Show that the following combinatorics identity is true:

$$\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}.$$ (4 marks)

(b) In how many ways can 3 men and 2 women sit in a row if the men and women are not to sit together? (4 marks)

A5. (a) The discrete random variable X assumes values 0 , 1 and 10 only. Given that

$$E(X)=2 \text{ and } Var(X)=13.$$

Determine the cumulative distribution function of X. (7 marks)

(b) The cumulative distribution function of a continuous random variable X is given by

$$f(x)=\begin{cases}12x^2(1-x) & , \quad 0\le x\le 1 \\ 0 & , \quad \text{Otherwise}\end{cases}$$

(i) Given that the random variable $Y=3X-4$, find the mean and variance of Y. (7 marks)

(ii) Find the probability that X lies between the mean and the mode.

(6 marks)

B2. (a) The random variable X has the exponential distribution with probability density function

$$f(x)=\begin{cases}\lambda e^{-\lambda x} & , \quad x>0\ ,\ \lambda>0 \\ 0 & , \quad \text{Otherwise}\end{cases}.$$

(i) Show that expectation of X is $E(X)=\dfrac{1}{\lambda}$ (4 Marks)

(ii) Show that $P(X>a)=e^{-\lambda a}$ for any $a>0$. Deduce the median of X.

(5 Marks)

(iii) For any $b>0$, find $P(X>a+b|X>a)$ and comment on this result.

(4 Marks)

(b) Now consider the case where $\lambda=1$.

(i) Sketch the graph of $f(x)$. (4 Marks)

(ii) State with a reason whether the distribution of X is positively or negatively skew. (2 Marks)

(iii) Write down the mode of the distribution of X, and find the value of k such that $Mean-Mode=k[Mean-Median]$. (3 Marks)

(iv) A student has read that, for many distributions,

- the skewness is positive if the mean is greater than the median
- the value of k is about 3.

Comment on the truth of each of these statements for the distribution of X.

(3 marks)

B4: (a) Write down the probability density function (pdf) of a continuous random variable X having the Normal distribution $N\left(\mu,\sigma^2\right)$ indicating the range of values of X and the parameter spaces for μ and σ^2. (5 Marks)

(b) State four properties of the Normal distribution. (2 marks)

(c) Show that the random variable $Z=\dfrac{X-\mu}{\sigma}\sim N(0,1)$ and write down the pdf of Z.

(5 Marks)

(d) A student arrives at a school X minutes after 08:00 A.M., where X may be assumed to be normally distributed. On a particular day it is observed that 40 % of the students arrive before 08:30 A.M. and 90 % arrive before 08:55 A.M.

(i) Find the mean and standard deviation of X. (7 Marks)

(ii) The school has 1,200 students and classes start at 09:00 A.M. Estimate the number of students who will be late on that day. (3 Marks)

(iii) Abebrese had not arrived by 08:30 A.M. Find the probability that he arrived late. (3 marks)